

TABLE 1
SEVERITY OF RISK FACTORS

Risk	Weight	Severity	Occurance	Detection
Very High	10	Results in non-compliance to government regulations, endanger life or machine	Almost certain failure	This failure will not be detected
	9		Well known history of failure from similar situation	Very difficult to detect
High	8	High degree of customer dissatisfaction. Causes problem in connected machines	Will fail occasionally due to this problem	Controls have poor chance of detecting this failure mode
	7			
Moderate	6	Customer dissatisfaction. Unscheduled repair. Damage to equipment.	Isolated case of failure from similar causes	Controls may detect the failure
	5			
	4			
Low	3	Will cause deterioration of performance and minor inconvenience	Very rare chance of failure happening	Controls automatically detect this failure and take preventive action
	2			
Minor	1	No noticable problem	Failure unlikely from this problem	

fying what can go wrong with the product. There can be various ways to identify these possible failure modes.

My favourite option is to sit together with the development team and examine each of the subsystems to find things that can go wrong. Once the team has listed all that can go wrong in each of the subsystems, we start examining things that can go wrong with interactions of the subsystems.

One should note that all problems are not equally important. Events that can cause life risk or big financial loss have to be given higher priority than a problem that will cause minor irritation. Similarly, problems that are very likely to happen need to be addressed with higher priority than a problem that has very rare chance of occurring. Defects that are difficult to identify need to be given higher priority than defects that are easy to identify.

Once we list the possible issues, we need to understand their effects, probability of such events occurring, and difficulty in detecting the issues. For each of the impact, probability, and detection we assign a subjective number according to the scale given in Table 1. Multiplication of these three numbers gives us the risk priority number (RPN).

Types of risk

Product risks can be of a technical nature, they can be economic, or they can be related to supply chain. Design deficiency, changes in input conditions, environmental factors, and the quality of input materials are technical risks.

Some of the technical risks can lead to accidents, where the impact can extend beyond the application area of the product. For this reason, technical risks need to be given higher priority in the analysis.

Economic risks cause loss of profitability or may cause higher operating cost. In both the cases business viability of the product suffers. Typical reasons for economic risks are price escalation, unplanned operating expenditure, or poor quality of supply. In some cases frequent breakdown due to technical fault may also result in economic risk.

The third kind of risk arises from supply-chain issues. Dependence on unreliable vendors, insufficient product volume, or product complexities often result in the suppliers not supplying the material. Supply-chain risks may call for costly alternatives, redesign of product, or cause closure of its production. Just like the product design, one has to be careful with vendor development or work with standard easily avail-

able raw material.

As a general guideline, any risk with RPN number higher than 400 is not acceptable. Such risks have to be reduced by redesign or by adding protective measures. One may ignore the risks with RPN value less than 120 and focus on reducing other higher priority risks. Risks with RPN value below 240 are somewhat tolerable.

Risks with RPN between 240 and 400 should be reduced. Such risks should be accepted only when the design team cannot come up with any feasible measure to reduce the risk.

Risk mitigation approaches

Risk mitigation can be done by various means. The aim of all the means is to reduce the RPN value. The best way for risk reduction is to either eliminate it or to reduce chance of it happening. This can be achieved by adding some safety margin in the design, by providing parallel sub-systems, or providing an auxiliary system that automatically takes over in the event of failure of the main system.

In addition to risk prevention measures, we can try to reduce the impact of the risk. We are all familiar with the fuse or circuit breakers in our domestic electric wiring. These fuses stop electric supply in